Pancreas

The second largest gland in the digestive system is the Pancreas, which is a yellow coloured, compound elongated organ consisting of exocrine and endocrine cells. It is situated between the limbs of the 'U' shaped duodenum. The exocrine portion secretes pancreatic juice containing enzymes such as pancreatic amylase, trypsin and pancreatic lipase and the endocrine part called Islets of Langerhans, secretes hormones such as insulin and glucagon. The pancreatic duct directly opens into the duodenum.

List the chemical preservatives, artificial enhancers found in the food items available in the market. How can you avoid such harmful substances in your food?

5.2 Digestion of food and role of digestive enzymes

The process of digestion converts the solid food into absorbable and assimilable forms. This is accomplished by mechanical and chemical processes.

Digestion in the buccal cavity

The smell, sight and taste as well as the mechanical stimulation of food in the mouth, triggers a reflex action which results in the secretion of saliva. The mechanical digestion starts in the mouth by grinding and chewing of food. It is called mastication. The saliva contain water, electrolytes (Na^+, K^+, Cl^-, HCO_3^-), salivary amylase (ptyalin), antibacterial agent lysozyme and a lubricating agent mucus (a glycoprotein). The mucus in saliva prepares the food for swallowing by moistening, softening, lubricating and adhering the masticated food into a bolus. About 30 percent of polysaccharide, starch is hydrolyzed by the salivary amylase enzyme into disaccharides (maltose). The **bolus** is then passed into the pharynx and then into the oesophagus by swallowing or **deglutition**. The bolus further passes down through the oesophagus to the stomach by successive waves of muscular contraction called **peristalsis**. The **gastro oesphageal sphincter** controls the passage of food into the stomach.

Digestion in the stomach

Food remains in the stomach for 4 to 5 hours, the rhythmic peristaltic movement churns and mixes the food with gastric juice and make it into a creamy liquid called **chyme**. The gastric secretion is partly controlled by autonomic reflexes. The secretion of gastric juice begins when the food is in the mouth. The gastric juice contains HCl and proenzymes. The proenzyme pepsinogen, on exposure to HCl gets converted into the active enzyme pepsin which converts proteins into proteoses and peptones (peptides). The HCl provides an acidic medium which is optimum for pepsin, kills bacteria and other harmful organisms and avoids putrifaction. The mucus and bicarbonates present in the gastric juice play an important role in lubrication and protection of the mucosal epithelium from the eroding nature of the highly acidic HCl (Figure. 5.8). Another proteolytic enzyme found in gastric juice of infants is rennin helps in the digestion of milk protein, caseinogen to casein in the presence of calcium ions. This enzyme secretion gradually reduces with aging.

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